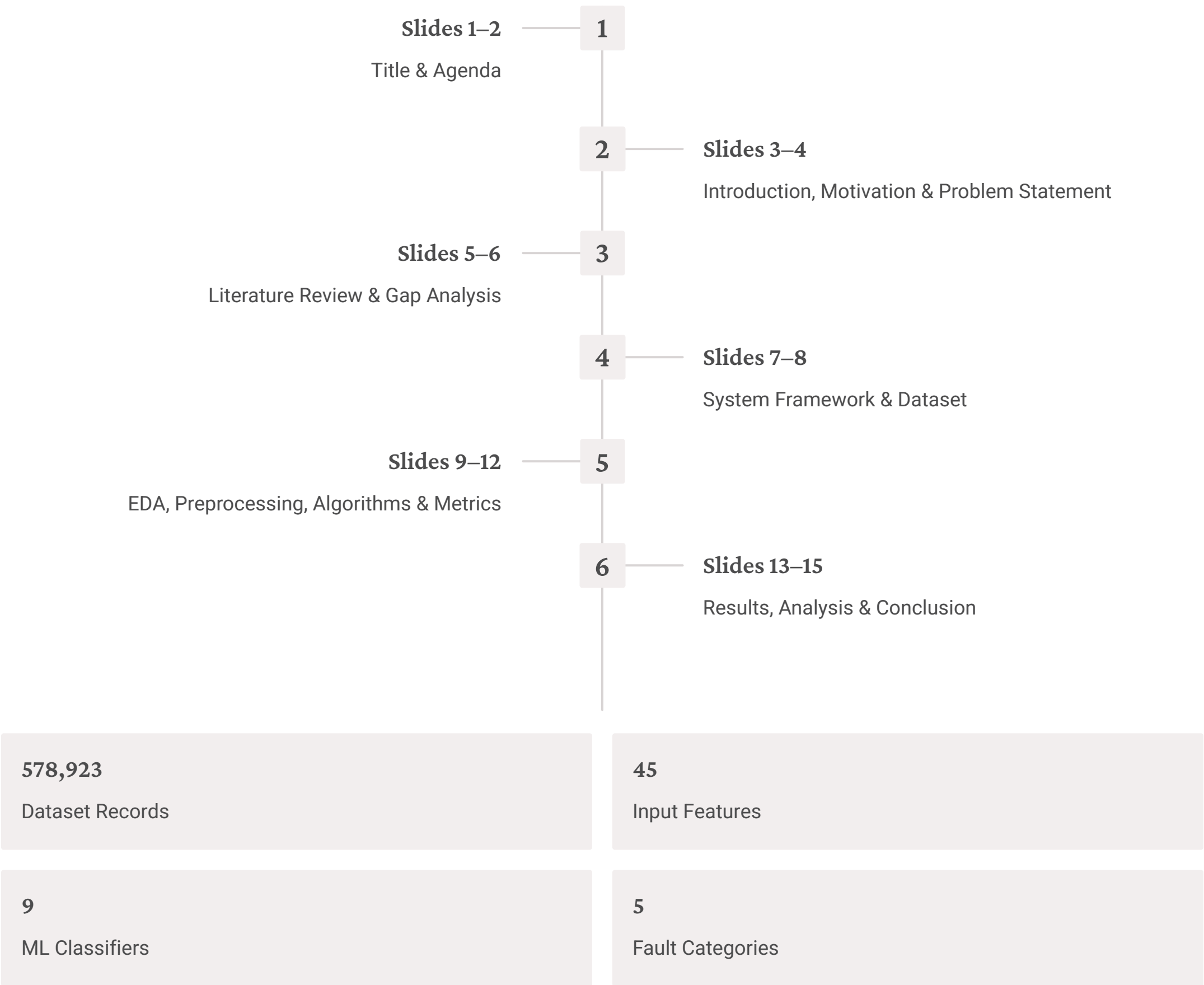


Presentation Structure

This presentation is organised into **15 slides** covering a complete, end-to-end machine learning project for transmission line fault detection.



Introduction & Motivation

The Critical Role of Power Transmission Lines

Power transmission lines form the **backbone of modern electrical infrastructure**. They carry electricity at high voltages across hundreds to thousands of kilometres — from generation stations to substations and ultimately to consumers. Any disruption in this network has cascading consequences for society and industry.

Environmental Threats

- **Atmospheric** — Lightning strikes, heavy rain, extreme winds, ice accumulation
- **Biological** — Bird contact, vegetation growth and tree interference
- **Pollution** — Industrial particulates and salt deposits degrading insulation
- **Aging** — Insulator wear, conductor fatigue, hardware corrosion

Fault Types & Severity

LG — Line-to-Ground

Single phase contacts earth — most common fault type

LL — Line-to-Line

Two phases make contact — moderate severity

LLG — Double Line-to-Ground

Two phases contact earth simultaneously

LLLG — Three-Phase

All three phases fault — most severe, least common

- ❏ **The Machine Learning Opportunity:** Modern smart grids produce continuous, multi-dimensional electrical data. ML can extract patterns to classify faults automatically — faster, more accurately, and more adaptively than any rule-based system.

Problem Statement & Objectives

The Core Problem

Existing fault detection systems are **reactive rather than intelligent**. They cannot:

- Distinguish between different fault types with high reliability
- Handle the scale of data from modern smart grid sensors
- Detect rare or intermittent faults appearing infrequently in operational data
- Adapt to changing grid topologies or seasonal environmental shifts

A critical technical challenge is **class imbalance**: the vast majority of readings represent normal "No Fault" operation. Naive models learn to ignore minority fault classes — achieving high paper accuracy while detecting no actual faults.

Project Objectives

01

Acquire and systematically explore the IEEE DataPort dataset (578,923 records, 45 features)

02

Conduct comprehensive EDA — visualise distributions, compute correlation matrices, identify informative predictors

03

Address severe class imbalance using **SMOTE** applied exclusively to training data

04

Implement and train **nine diverse ML classifiers** spanning linear, tree-based, ensemble, kernel, and neural families

05

Rigorously evaluate all models using accuracy, precision, recall, F1-score, and confusion matrices

06

Identify the best-performing model and provide actionable deployment recommendations

Literature Review — 10 Key Research Works

1	ANN-Based Fault Detection <i>Abniki, Razavi & Nateghi (2012)</i> – ANN with 6 inputs and 11 output nodes for T-connected overhead lines. Established ANN as a viable approach for single-phase and multi-phase fault classification.
2	Summation of Squared Currents <i>Yamuna & Thresia (2022)</i> – Combines three phase current signals into a single SSC scalar, reducing complexity while retaining discriminatory power for resource-constrained deployment.
3	CNN for IEEE 9-Bus System <i>Jiriwibhakorn & Kanwal (2025)</i> – CNNs applied to time-series fault waveforms, automatically extracting hierarchical features – eliminating handcrafted feature engineering.
4	Alienation Coefficients + SVM <i>Huang (2023)</i> – Waveforms processed into alienation coefficients fed to SVM. Demonstrated compact derived features yield strong classification performance.
5	GRNN + SVD for HVDC <i>Cui & Tu (2015)</i> – Generalised Regression Neural Network with SVD for HVDC systems. SVD reduces dimensionality while preserving fault-relevant information.
1	Federated Learning + Blockchain <i>Mukisa et al. (2025)</i> – Privacy-preserving framework for smart grids. Distributed substations train a shared model without sharing raw data; blockchain provides tamper-proof audit trails.
2	IoT-Based Real-Time Monitoring <i>Mohan et al. (2024)</i> – IoT sensors deployed along transmission corridors capturing phase voltages and currents, with automated threshold-based alerting in near-real-time.
3	MATLAB/Simulink Simulation <i>Sahu et al. (2025)</i> – Systematic simulations for LG, LL, LLG, and LLLG faults under varying resistance and location conditions. Provides a structured performance benchmark.
4	AI in Smart Grids — Survey <i>Nayyer & Nayyer (2024)</i> – Broad survey of AI applications across smart grid domains. Highlights rapid ML adoption and identifies challenges including interpretability and real-time deployment.
5	Environmental Factor Modelling <i>Zhiqiang et al. (2020)</i> – Probabilistic models linking lightning discharge, pollution severity, ice load, and vegetation encroachment to fault risk and waveform characteristics.

Gap Analysis & Research Justification

A systematic review of the 10 prior works reveals **five persistent gaps** that this project directly addresses:



Gap 1 — Class Imbalance Neglected

Fault events are inherently rare. Most studies do not account for imbalance — models develop majority-class bias, achieving high accuracy while failing to detect actual faults.



Gap 2 — Isolated Algorithm Studies

Each paper evaluates one or two algorithms. No existing study provides a comprehensive, controlled head-to-head comparison across diverse algorithm families on identical data.



Gap 3 — Simulated Data Dominance

Many studies use MATLAB/Simulink synthetic data. Simulated environments do not capture real-world noise, sensor drift, electromagnetic interference, or irregular environmental fault patterns.



Gap 4 — Incomplete Evaluation

Many studies report accuracy as the sole metric. On imbalanced datasets, accuracy is misleading. Recall, precision, and per-class F1-scores are essential for safety-critical assessment.



Gap 5 — Data Security & Privacy

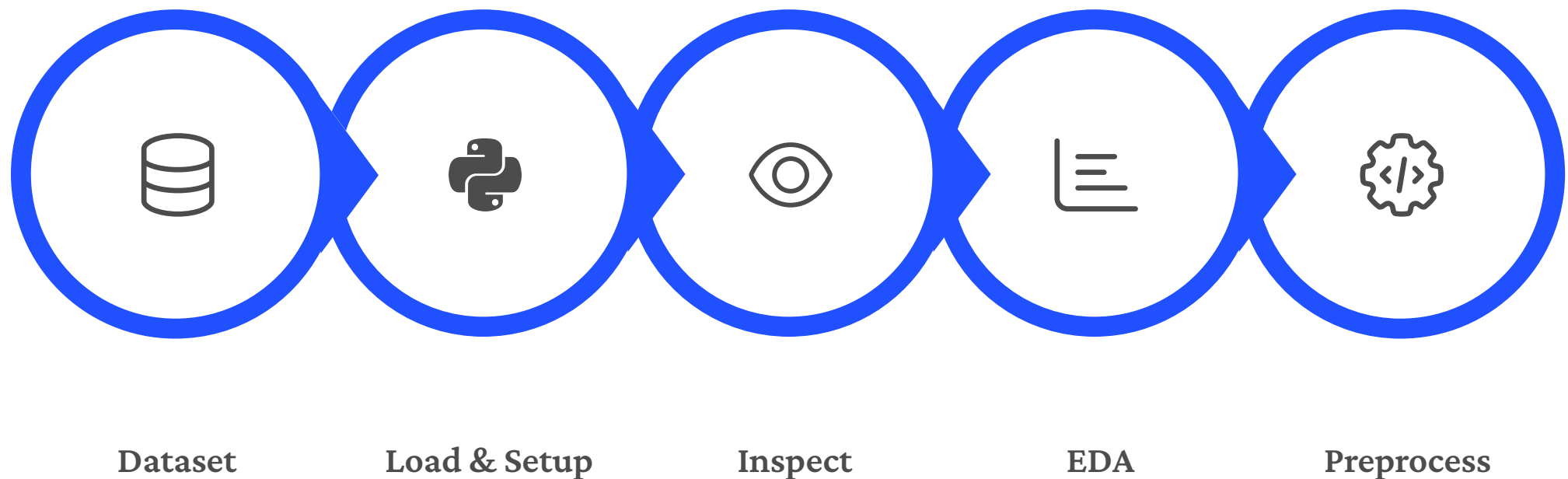
Centralised ML architectures require aggregating sensor data — creating significant security vulnerabilities. Privacy-preserving alternatives remain largely unexplored in fault detection literature.

How This Project Addresses These Gaps

Gap	This Project's Response
Class Imbalance	SMOTE applied to training data before all model training
Limited Comparison	9 algorithms across 6 families benchmarked on identical dataset
Simulated Data	Real-world IEEE DataPort dataset — 578,923 sensor records
Incomplete Evaluation	Accuracy + Precision + Recall + F1 + Confusion Matrix for all models
Security	Federated learning identified as primary future extension

Proposed System Framework & Pipeline

The proposed system processes raw transmission line electrical measurements through **eight structured stages**, transforming sensor data into reliable fault classifications.



Each stage is carefully sequenced to ensure data integrity – notably, SMOTE is applied **exclusively to the training partition** to prevent information leakage into the test set, and StandardScaler is fitted only on training data before transforming both splits.

Tools & Libraries

- Python – Google Colab environment
- Pandas & NumPy – data handling
- Scikit-learn – modelling & evaluation
- Matplotlib & Seaborn – visualisation

Key Design Decisions

- Stratified 80/20 train-test split
- SMOTE on training data only
- StandardScaler for distance-sensitive models

Dataset Description

IEEE DataPort — Transmission Line Fault Using Line Voltages and Currents Features

578K

Total Records

578,923 sensor readings from
IEEE DataPort

45

Total Attributes

43 numeric inputs + 2
categorical targets

0%

Missing Values

Complete dataset — no
imputation required

5

Fault Classes

No Fault, LG, LL, LLG, LLLG

Input Features (43 Numeric)

Phase Voltages — Va, Vb, Vc

Instantaneous voltage measurements on each of the three phases, recorded at multiple monitoring points along the transmission line.

Phase Currents — Ia, Ib, Ic

Instantaneous current measurements on each phase — providing the primary fault-discriminating signals for classification.

Derived Parameters

Additional computed quantities from voltage-current relationships across different sensor positions, capturing line impedance characteristics.

Target Variables

Target 1 — Fault Type

- **No Fault** — Normal operating condition (dominant majority class)
- **LG** — Line-to-Ground: single phase contacts earth
- **LL** — Line-to-Line: two phases make electrical contact
- **LLG** — Double Line-to-Ground: two phases contact earth
- **LLL** — Three-Phase: all phases involved — most severe

Target 2 — Fault Location

Identifies the precise position along the transmission line where the fault has occurred. Enables maintenance teams to dispatch directly to the fault site without time-consuming manual inspection traversals.

📄 **Data Quality:** The dataset is complete with no missing values across all 578,923 records. Its large scale provides excellent statistical coverage for training robust ML models capable of generalising to unseen fault conditions.

Exploratory Data Analysis (EDA)

Understanding Data Characteristics Before Modelling

Exploratory Data Analysis was conducted before any modelling to ensure that design decisions – preprocessing steps, algorithm selection, and evaluation strategy – are grounded in the actual properties of the data.

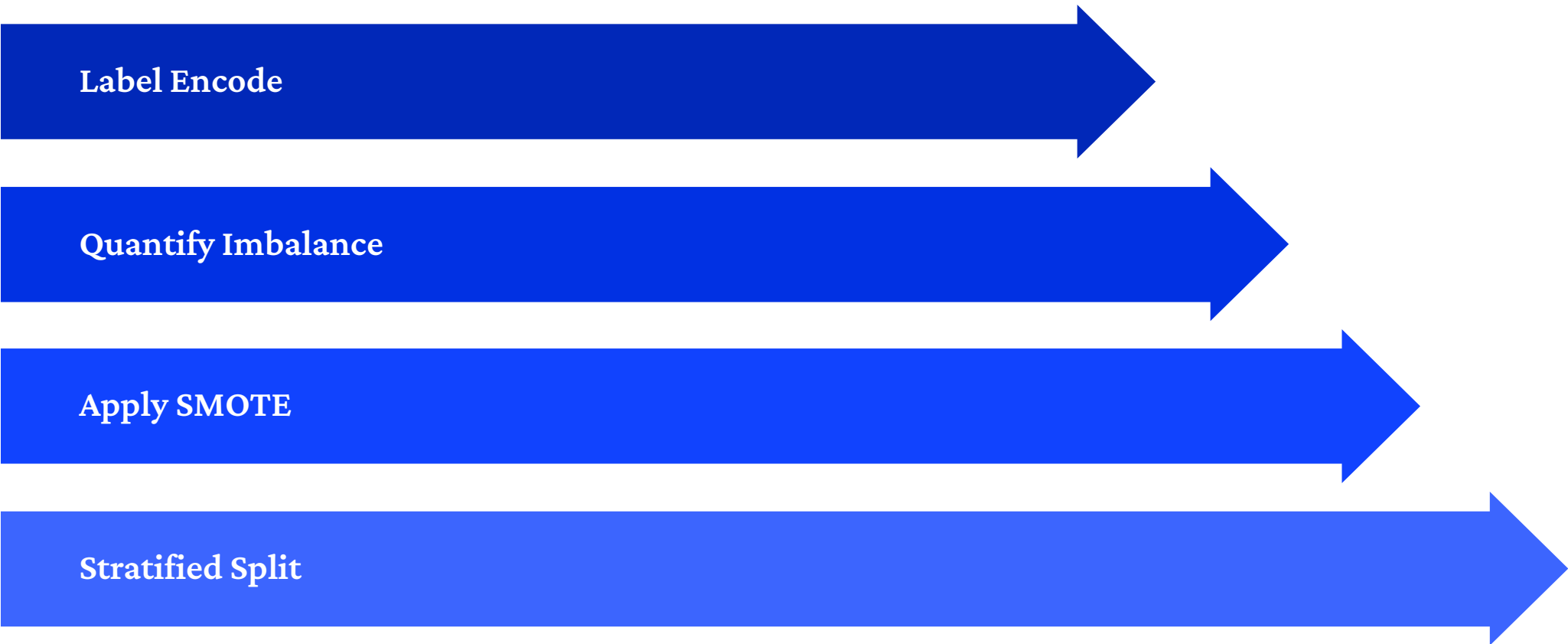
1 Severe Class Imbalance The "No Fault" class constitutes the overwhelming majority of records. The four fault classes (LG, LL, LLG, LLLG) together represent a small fraction of the total dataset. - A naive classifier predicting "No Fault" achieves over 80% accuracy yet detects zero actual faults - This makes SMOTE not optional, but essential	2 Feature Correlation (All 45 Features) A full 45×45 correlation heatmap was generated. Features derived from the same phase exhibit strong positive correlation due to Ohm's law. Cross-phase correlations exist due to electromagnetic mutual inductance. Several feature clusters show high internal correlation, suggesting potential redundancy
3 Top 10 Predictive Features Filtering to the top 10 features most correlated with the fault-type target reveals key predictors. - Phase current measurements (Ia, Ib, Ic) are the strongest predictors - Voltage features carry predictive information but are secondary to currents - Feature combination matters more than any single measurement	4 Distinct Fault Signatures Visual inspection of feature distributions across fault classes confirms that different fault types produce statistically distinct electrical signatures – validating the core hypothesis that pattern-based ML classification is theoretically sound.

❑ The classes are separable – the challenge is building a model sensitive enough to identify rare fault patterns against the dominant No-Fault background.

Data Preprocessing & SMOTE

Preparing Data for Reliable Machine Learning

Five preprocessing steps were performed sequentially to transform raw data into a form suitable for training rigorous, unbiased machine learning models.



Steps 1–3: Encoding & Balancing

Label Encoding

Categorical fault labels converted to integers: No Fault → 0, LG → 1, LL → 2, LLG → 3, LLLG → 4. LabelEncoder from Scikit-learn applied consistently to both target columns.

Imbalance Quantification

The exact class ratio is computed before any correction. The severe imbalance is confirmed numerically — not merely assumed — justifying the SMOTE intervention.

SMOTE Application

Synthetic Minority Oversampling applied **only to the training partition** — never to the test set. Prevents data leakage. After SMOTE: all five fault classes represented with equal frequency in training data.

Steps 4–5: Splitting & Scaling

Stratified Train-Test Split

80% training data (SMOTE applied post-split) and 20% held-out test data (original distribution, unmodified). Stratified splitting preserves class proportions. Random state fixed for full experimental reproducibility.

Feature Scaling

Tree-based models are inherently scale-invariant — no scaling required. Distance-sensitive models (SVM with RBF kernel, KNN) require StandardScaler: features normalised to zero mean and unit variance.

 **Critical Detail:** SMOTE is applied only to the training partition. The test set preserves the original real-world class distribution to ensure evaluation reflects realistic deployment conditions.

Machine Learning Algorithms

Nine Classifiers Across Six Algorithm Families

Family 1 — Linear Baseline

Logistic Regression: Linear decision boundary via sigmoid. Fast, interpretable. Serves as experimental baseline. Applied with `class_weight='balanced'`.

LinearSVC: Linear kernel SVM optimised with hinge loss. Efficient on high-dimensional data. Shares the linear boundary limitation.

Family 2 — Tree-Based

Decision Tree: Recursively partitions feature space using threshold conditions. Non-linear boundaries achievable. Fully human-readable and auditable — the only model offering complete interpretability alongside high performance.

Family 3 — Ensemble

Random Forest: Bagging of independent trees on random feature subsets. Reduces variance and overfitting.

AdaBoost: Sequential boosting — each learner focuses on previously misclassified samples. Effective on rare minority classes.

Gradient Boosting: Trees minimise residual error via gradient descent. Typically most accurate on tabular data.

Family 4–6 — Kernel, Instance & Probabilistic

SVM (RBF): Maps features to infinite-dimensional space. Highly expressive but computationally expensive.

KNN: No training phase — classifies by K nearest neighbours. Flexible but $O(n)$ prediction cost.

Naïve Bayes: Bayes' theorem with independence assumption — violated by correlated electrical measurements.

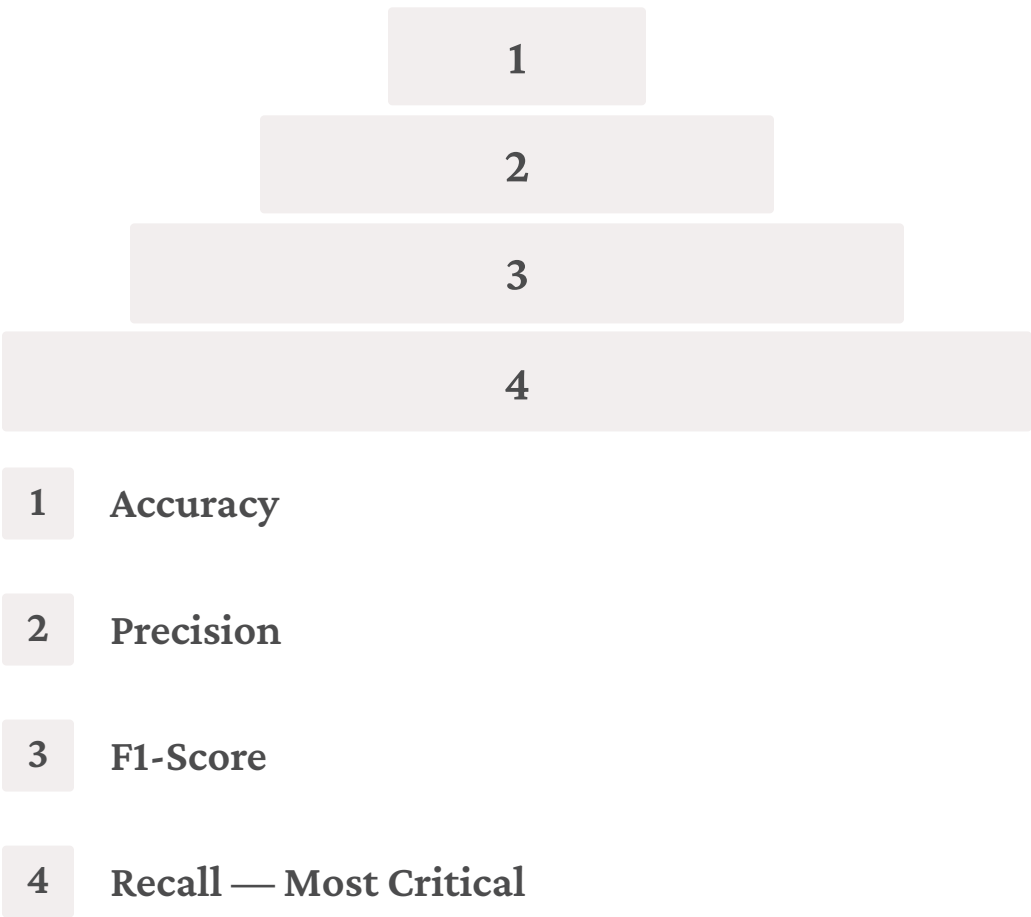
MLP Classifier: Feedforward neural network via backpropagation. Learns complex hierarchical representations.

Evaluation Metrics & Methodology

A Rigorous, Multi-Metric Evaluation Framework

The Accuracy Trap: A classifier predicting "No Fault" for every input achieves ~80%+ accuracy — yet detects zero actual faults. This illustrates precisely why accuracy-only evaluation is inadequate for imbalanced fault detection tasks.

Metric Priority Hierarchy



For power system fault detection: **Recall > F1-Score > Precision > Accuracy**. Undetected faults risk equipment damage, grid instability, and physical danger.

Accuracy

$(TP + TN) / (TP + TN + FP + FN)$ — Fraction of all correct predictions. Used for cross-model comparison but never interpreted in isolation.

Precision

$TP / (TP + FP)$ — Of all samples classified as a fault type, what fraction actually belong to that class? Minimises false alarms and unnecessary maintenance dispatches.

Recall (Sensitivity)

$TP / (TP + FN)$ — Of all actual fault occurrences, what fraction does the model detect? **Most critical metric for power system safety.** A missed fault can escalate into catastrophic failure.

F1-Score

$2 \times (Precision \times Recall) / (Precision + Recall)$ — Harmonic mean of Precision and Recall. Penalises models that sacrifice either dimension excessively. Primary comparative metric.

Confusion Matrix

Per-class breakdown of TP, TN, FP, FN for every fault category. Reveals which specific fault pairs the model confuses — essential for diagnosing weaknesses.

Experimental Results

Comparative Model Performance — Held-Out Test Set

All models evaluated on the same 20% held-out test set. Test data was never resampled — it preserves the original real-world class distribution for an honest evaluation.

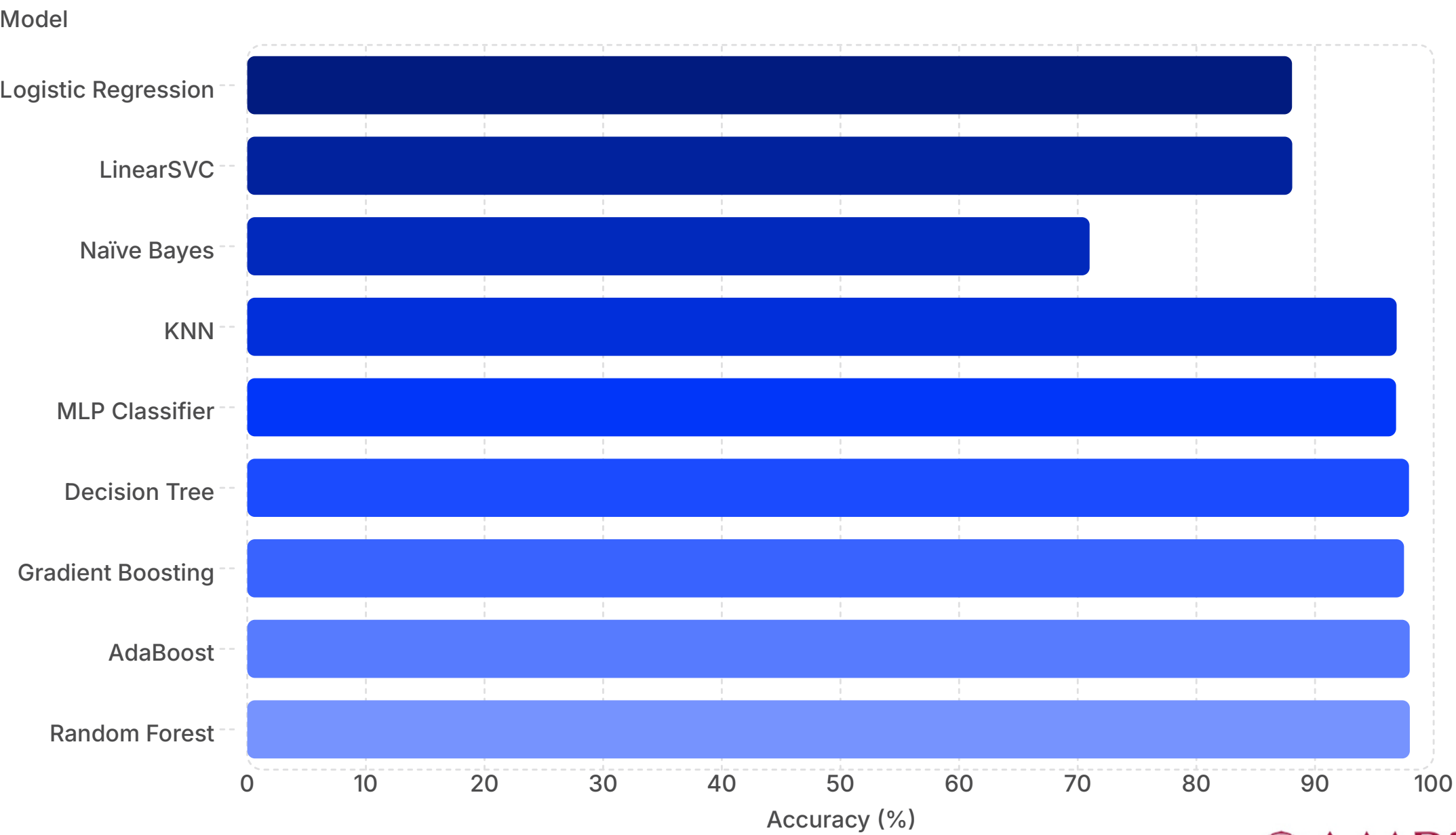
Model	Accuracy	Performance on Fault Classes	Verdict
Logistic Regression (no SMOTE)	88.06%	Very low recall on all fault types — majority class bias	Poor
Logistic Regression (with SMOTE)	31.40%	Recall improved but precision collapsed completely	Poor
LinearSVC	88.08%	Same fundamental limitation as Logistic Regression	Poor
Naïve Bayes	66.23%	Moderate recall, low precision — independence assumption violated	Poor
KNN	96.88%	High precision and recall across all classes	Good — slow inference
MLP Classifier	96.83%	Strong overall, some instability on minority classes	Good — tuning needed
Decision Tree	97.92%	Very high precision, recall, F1 — fully interpretable	Excellent
Gradient Boosting	95.40%	Very high across all metrics — strong ensemble	Excellent
AdaBoost	97.98%	Highest minority-class recall — boosting advantage	Excellent
Random Forest	97.99%	Highest precision, recall, and F1 across all classes	Best — Recommended

The Linear-vs-Ensemble Performance Gap

The ~10 percentage-point accuracy gap between linear models (~88%) and ensemble models (~98%) directly reflects the inherent non-linearity of electrical fault signatures in three-phase transmission systems. Linear classifiers are architecturally incapable of capturing these patterns regardless of tuning.

The SMOTE Effect — A Nuanced Interaction

SMOTE interacts fundamentally differently with different model families. On Logistic Regression: accuracy dropped from 88% to 31% — SMOTE shifted the linear boundary too aggressively. On ensemble models: accuracy remained high while recall on minority fault classes improved — the ensemble architecture absorbs SMOTE's rebalancing without boundary collapse.



Conclusion, Limitations & Future Work

Summary of Findings

This project successfully demonstrated that machine learning can detect and classify faults in power transmission lines with high accuracy, precision, and recall – using only standard voltage and current sensor measurements. Nine algorithms were rigorously benchmarked on a large, real-world dataset (578,923 records).

97.99%	97.98%	97.92%	578,923
Random Forest	AdaBoost	Decision Tree	Records Benchmarked
Top performer – highest precision, recall, and F1 across all five fault classes	Near-identical performance with superior minority-class recall in some configurations	Best balance of high accuracy and complete interpretability for regulated engineering environments	IEEE DataPort dataset – Transmission Line Fault Using Line Voltages and Currents Features

Limitations of This Study

- Dataset from IEEE research repository – not a live operational power grid; real-world generalisation remains to be validated
- Hyperparameter tuning performed through manual experimentation rather than exhaustive systematic search
- No deployment, embedded hardware integration, or real-time latency testing conducted
- Models not tested under adversarial conditions: sensor failures, communication dropouts, or extreme environmental noise

Future Research Directions

01

Deep Learning on Time-Series Waveforms

Apply LSTM or 1D-CNN architectures to raw multi-channel voltage and current waveforms to capture temporal fault dynamics

02

Real-Time Fault Monitoring Pipeline

IoT sensor streams feeding a continuously running inference engine – enabling sub-second fault classification and automated alert dispatch

03

Federated Learning Deployment

Distribute model training across multiple substations without centralising raw sensor data – preserving privacy and eliminating single-point-of-failure risks

04

Automated Feature Selection via SHAP

Quantify each feature's contribution to fault classification decisions – reducing model complexity and inference cost

05

Multi-Task Joint Prediction

Unified model simultaneously predicting fault type and fault location as joint outputs in a single inference call

Appendix A — Dataset Sample & Feature Reference

Table A.1 — Example Records from the Dataset

The following table illustrates a representative sample of the dataset structure, showing how electrical measurements map to fault categories. Each row represents one sensor snapshot from the transmission line monitoring system.

Record #	Va	Vb	Vc	Ia	Ib	Ic	Type of Fault	Fault Location
1	230.1	229.8	230.0	5.2	5.1	5.2	No Fault	—
2	198.4	231.0	230.1	18.7	5.0	5.1	LG (Line-to-Ground)	45 km
3	201.2	201.5	229.9	14.3	14.1	5.0	LL (Line-to-Line)	82 km
4	185.3	188.1	230.2	21.4	20.8	5.2	LLG (Double Line-to-Ground)	113 km
5	190.0	191.2	189.7	22.1	21.9	22.3	LLLG (Three-Phase)	67 km

Feature Description — All 45 Attributes

Phase Voltages

Va, Vb, Vc (at multiple sensor nodes)

Numeric (float) — **Input**

Phase Currents

Ia, Ib, Ic (at multiple sensor nodes)

Numeric (float) — **Input**

Derived Parameters

Impedance ratios, power factor estimates

Numeric (float) — **Input**

Type of Fault

LG / LL / LLG / LLLG / No Fault

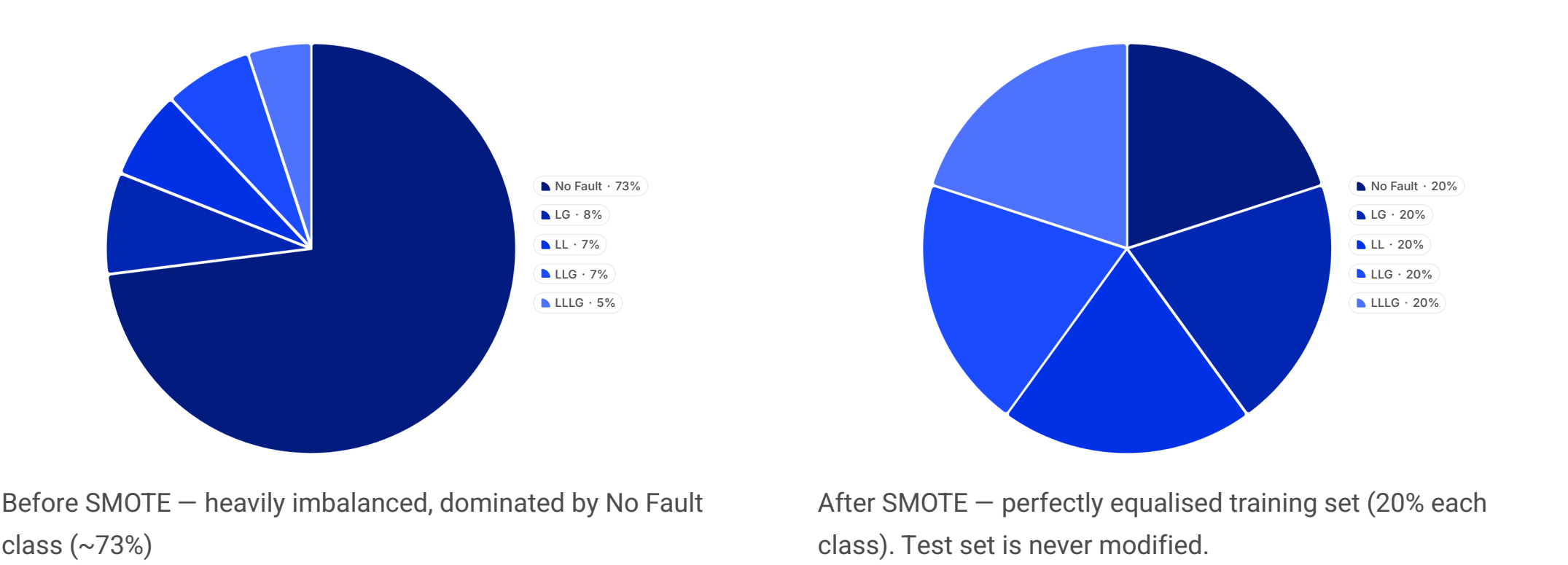
Categorical → int — **Target 1**

Fault Location

Distance along line in km

Categorical → int — **Target 2**

Class Distribution Before and After SMOTE



Appendix B & C — Software Environment & Mathematical Foundations

Software Tools & Environment

Tool / Library	Version	Purpose
Python	3.10+	Primary programming language
Google Colab	Cloud	Development & execution environment (free GPU)
NumPy	1.24+	Numerical array computation and linear algebra
Pandas	2.0+	Dataset loading, manipulation, and inspection
Matplotlib	3.7+	Static data visualisation (distribution plots, confusion matrices)
Seaborn	0.12+	Statistical visualisation (heatmaps, pairplots)
Scikit-learn	1.3+	ML model implementation, preprocessing, and evaluation
imbalanced-learn	0.11+	SMOTE implementation for class balancing
Joblib	1.3+	Model serialisation and parallel processing

Why Google Colab?

- Zero local setup – accessible from any browser
- Free GPU/TPU compute (T4 GPU, 15 GB VRAM) for large dataset training
- Seamless Google Drive integration for dataset persistence
- Pre-installed ML libraries eliminating dependency management overhead
- Shareable notebooks enabling collaborative development across the team

Mathematical Foundations

C.1 — Evaluation Metric Formulas

Let TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives. For multi-class classification, each metric is computed per class (one-vs-rest) and macro-averaged:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$
$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$
$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$
$$\text{Macro-}F_1 = \frac{1}{n} \sum_{i=1}^n F_{1,i}$$

C.2 — SMOTE Synthetic Sample Generation

$$x_{\text{new}} = x_i + \lambda \times (x_j - x_i), \quad \lambda \sim \text{Uniform}(0, 1)$$

Generates samples along the line segment connecting real minority samples – producing realistic interpolated instances rather than duplicates or noise.

C.3 — Random Forest Decision Rule

$$\hat{y} = \arg \max_c \sum_{i=1}^D \mathbf{1}[T_i(x) = c]$$

Each of D trees votes; the majority class wins.
`class_weight='balanced'` scales contributions by $n_{\text{samples}} / (n_{\text{classes}} \times \text{class_count}[y])$.

C.4 — AdaBoost Update Rule

$$\varepsilon_t = \sum_i w_i \cdot \mathbf{1}[y_i \neq h_t(x_i)]$$
$$\alpha_t = 0.5 \times \ln \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)$$
$$w_i \leftarrow \frac{w_i \cdot \exp(-\alpha_t \cdot y_i \cdot h_t(x_i))}{\sum_j w_j}$$
$$H(x) = \text{sign} \left(\sum_t \alpha_t \cdot h_t(x) \right)$$

Appendix D — Full References

Complete Bibliography — IEEE Format

[1] Abniki et al., 2012 H. Abniki, M. N. Razavi, and S. Nateghi, "An ANN-Impedance-based Fault Detection Technique in T-Connection Transmission Overhead lines," in <i>Proc. IEEE Electrical Engineering and Computing Science (EECS)</i>, June 2012.	[2] Yamuna & Thresia, 2022 U. Yamuna and M. M. Thresia, "Fault Detection In Transmission Lines using Summation of Squared Three-Phase Currents," in <i>2022 International Conference on Futuristic Technologies in Control Systems & Wireless Communication (FTCSS)</i>, 2022, pp. 1–6.
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Dataset Citation

IEEE DataPort, "Transmission Line Fault Using Line Voltages and Currents Features," IEEE DataPort Repository, 2023. [Online]. Available: <https://ieee-dataport.org>

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